Who produces and who consumes knowledge on Wikipedia?

Abstract

Wikipedia is the biggest online encyclopedia in the world written exclusively by volunteers in over 300 different languages. However, articles are mostly written by individuals from developed countries; smaller countries aren't only underrepresented, but also rarely define and represent their own places themselves. This project aims to explore the dynamics of Wikipedia's knowledge ecosystem by focusing on how demographic and socioeconomic factors affect the production and consumption patterns of a user. The report discusses the relevant literature, discussing the motivations and geographic distributions of readers and editors, and bias in knowledge production and representation. Lastly, the report includes the project specification, defining the scope, methodology, deliverables, timeline, and evaluation criteria.

☐ **I have not used any GenAI tools in preparing this assessment.**

☐ **I certify that all material in this dissertation which is not my own has been identified.**

1 Introduction

Written explicitly by volunteers, Wikipedia has been one of the most visited websites for over two decades [1]. The website was created in 2001 by two co-founders, Larry Sanger and Jimmy Wales, as a free, collaborative web encyclopedia [2]. Initially, Wikipedia was available only in English; however, nowadays articles can be found in more than 300 different languages, written by people from all over the globe. In [3], Marc M. Ribé mentions how Wikipedia is a website where you can find the "sum of all human knowledge" due to the large number of various topics available, ranging from mathematical articles to philosophy and thinking. In other words, anything that one can think of can probably be found on Wikipedia.

Despite Wikipedia being such a large platform for knowledge and information, not everyone is willing to dedicate time and effort to edit articles. In fact, only a minor portion of the Wikipedia users edit the articles, which is often called the 'participation inequality' [4] [5]. Moreover, the readers, otherwise known as 'non-contributors', are considered to be "lurkers", parasites who make little to no contribution to the encyclopedia, but benefit from others' work [3]. This attitude towards the readers has made it more challenging to become an editor, for readers are seen as a nuisance, rather than potential future contributors.

The majority of the Wikipedia editors and readers reside in English Wikipedia, which to this day produces the most number of articles. As of November 2024, there are around 6,920,000 articles in the English Wikipedia, compared to approximately 4,350,000 articles in February 2013 [6]. Pablo Beytía [7] states that most of the content created is related to the United States of America and Western Europe, as opposed to relatively smaller information about regions of Africa, Asia, the Middle East and Latin America. The spatial distribution of geo-tagged articles and distribution of subgroups of articles reveal that this trend is quite common, and can be found in biographies of well-known people, historical events and in documentation of animals, originating from a specific territory. Hence, it is evident that the more economically developed a country is, the more likely are the residents to edit and read Wikipedia [8].

Previous studies have focused on language bias, geographic distribution and editors and readers separately, usually targeting only a few countries or a specific region. They have shown that wealthier nations with better internet connections and education systems tend to dominate both content creation and consumption on Wikipedia. However, not much attention has been paid to low-income countries or countries that have a very low percentage of participation in Wikipedia. Furthermore, questions such as "Who's writing about small towns in Brazil?" or "Who reads and who writes the Arabic Wikipedia?" remain unanswered. This project aims to analyse the patterns of knowledge production and consumption on Wikipedia, examining geographic, cultural and socioeconomic factors that influence who contributes and who consumes content on the platform. The main goal is to use data on user edits and page views provided by Wikipedia, along with methods such as data mining, machine learning, and spatial data science, to uncover disparities in representation, assess biases in content coverage, and contribute to the fields of computational social science, inequality, and information geography.

2 Literature Review

2.1 Overview of Readers and Editors in Wikipedia

The Wikipedia community tends to split the contribution behaviour into two categories: those who participate by editing and those who read. Since it is a free online encyclopedia written by volunteers, many people benefit from the work of editors, in other words, "free-ride" [9]. This term implies that a reader makes a conscious choice of not contributing to Wikipedia articles. However, not all readers can be considered free-riders. For example, unawareness of how Wikipedia works can prevent an individual from fixing an error they noticed in an article [9].

In [10], Lemmerich et al. conduct a comparative study to reveal why individuals read Wikipedia in the first place. This is done by combining a large-scale survey, conducted in June 2017, in 14 different Wikipedia languages with a log-based analysis of user activity. The survey results show that the main motivation of reading in all languages, except for English, Dutch, and Japanese, is intrinsic learning. This means that most of the readers gain information from Wikipedia at their own will, fueled by their passion to learn new things and broaden their knowledge base. The three aforementioned countries, however, are motivated mainly by media. Other motivators include conversations, work or school-related tasks, current events, the need for making personal decisions, and boredom [10]. These findings indicate that there is no single dominating motivator. Another study [11] confirms this, stating that Wikipedia is usually used for checking facts and verifying them, rather than thoroughly studying a topic. Moreover, more often than not readers already know or have an idea about the topic beforehand.

The minority of people who are responsible for editing Wikipedia are commonly known as Wikipedians within the Wikipedia community [12]. In their research [13], Forte and Bruckman interview with 22 encyclopedia writers to get an answer to the question “Why do they do it?”. At first it seems that the contributors to the platform receive no credit for their hard work, for none of the articles are signed, and lots of articles have been countlessly edited by different people. During the interviews it was revealed that Wikipedia contributors actually recognise each other, which allows them to claim ownership of their work. For example, some of the interviewees maintain extensive watchlists and monitor both their articles and the ones they have significantly contributed to. Since the editing history is available for every article, it is possible to identify to who is the actual owner of an article [13]. Furthermore, Wikipedians have userpages, where they can claim ownership of their articles and put up a “resume” for themselves.

Naturally, editors do not contribute to Wikipedia with no purpose either. They are also motivated by specific factors, which are studied by Yang and Lai in [14]. They propose the four following hypotheses:

- “Hypothesis 1. Intrinsic motivation positively affects individual knowledge sharing behaviour in Wikipedia.”
- “Hypothesis 2. Extrinsic motivation positively affects individual knowledge sharing behaviour in Wikipedia.”
- “Hypothesis 3. External self-concept motivation positively affects individual knowledge sharing behaviour in Wikipedia.”
- “Hypothesis 4. Internal self-concept motivation positively affects individual knowledge sharing behavior in Wikipedia.”

The definition for intrinsic motivation is the same for editors as it is for readers, while extrinsic motivation is its complete opposite—It is a goal-oriented motivation which is pursued with the intent of obtaining a “reward” at the end [14] [15] [16]. External self-concept and internal self-concept motivations are similar, except one targets the expectations of a specific group, while the other targets an individual’s own standards, respectively [16]. According to the authors, the main motivator for most of the editors is intrinsic motivation, which had a mean score of 4.46. This was followed by internal self-concept motivation (3.64), extrinsic motivation (2.99) and external self-concept motivation (2.76).

Out of all the four hypotheses, only Hypothesis 4 was supported in the conducted study of the authors. The positive association between internal self-concept motivation and knowledge-sharing behaviour showed that editors gain confidence by contributing to Wikipedia. In other words, self-assurance is positively tied to knowledge-sharing behaviour in Wikipedia [14]. Individuals who are motivated by their internal self-concept are confident in demonstrating their abilities, which drives them to post more frequently on the platform. Moreover, intrinsic motivation does not really affect the knowledge-sharing behaviour of the editors. Yang and Lai mention that “enjoyment is gained in the process of sharing knowledge", explaining why Hypothesis 1 was unsupported in their study. Hypothesis 2 was unsupported because no reward given by Wikipedia is comparable to a physical reward. Extrinsic motivation, therefore, doesn't have a great impact on individual knowledge-sharing behaviour of Wikipedia users [14]. The same could be said about Hypothesis 3— Wikipedians are not related to any reference groups, for they have their own accounts, and none of the accounts are related to their physical identity. Altogether, both studies show that the biggest motivator for both readers and editors is intrinsic motivation, which elaborates on the curiosity and intelligence of the Wikipedia community.

2.2 Geographic Distribution of Wikipedia Readers and Editors

As outlined in the introduction, the geographics of participation to Wikipedia are unequal, and the biggest share of contributions come from a limited number of regions in the world. In [6], Graham et al. examined the geographic distribution of edits committed to all Wikipedias by using Wikimedia API. The analysis of the geography of participation revealed that the largest numbers of edits belonged to mostly economically developed countries situated in North America, Europe, and Asia, such as the United States of America, Germany, Japan or Canada (Figure 1). The rest of the countries with the most edits were in Europe, except six which were in Latin America and the Caribbean (Brazil, Mexico, Argentina, Chile, Colombia, and Peru) and two in Middle-East (Israel and Iran). The authors stated that the geography of participation in Wikipedia can be explained by a set of national-level indicators:

- Population
- GDP (Gross Domestic Product)
- GER (Gross Enrollment Ratio)
- Broadband Internet Connections

Population serves as a baseline variable linked to the potential pool of individuals who may participate in editing Wikipedia. GDP, which is commonly used to measure a country's total economic output, can represent various essential factors that facilitate Wikipedia editorship, including leisure time, technology access, education or libraries. GER measures the proportion of a country's population aged 5 to 17 that is enrolled to school. By using GER it is possible to establish basic literacy and educational levels needed to engage with text-based resources like Wikipedia. Lastly, Broadband Internet Connection makes it possible for individuals to edit Wikipedia. Graham et al. [6] state that it is likely that faster internet connections have a more positive impact on content creation compared to the slower, non-broadband connections.

Moreover, using these indicators in the regression models, authors reveal that the total number of edits significantly increases with the number of broadband internet connections in a country. The analysis showed that countries with either very low or very high broadband access seem to contribute more Wikipedia edits than a linear model would predict, contradicting the direct proportionality of internet access to the edits. For example, countries with medium access contribute way less than expected, suggesting that those who do have internet access in poor countries are active Wikipedians. It becomes evident that connectivity is the strongest predictor of participation in Wikipedia, rather than education, wealth or the population.

Another study [17] further investigated the economic imbalance by focusing on averaged monthly Wikipedia clicks per internet user (2011 to 2017). According to the authors, most of the Wikipedia clicks per internet user are highest for high-income countries and lowest for the poor countries, especially in Africa. The countries with distorted access to internet, such as China and Iran, are two major outliers. Despite having over half a billion internet users, China averages on only 0.2 monthly clicks per user, while for Iran it is 10 monthly clicks per user. The authors added that the usage of Wikipedia "does not only reflect the economic circumstances, but that it is rather related to differences

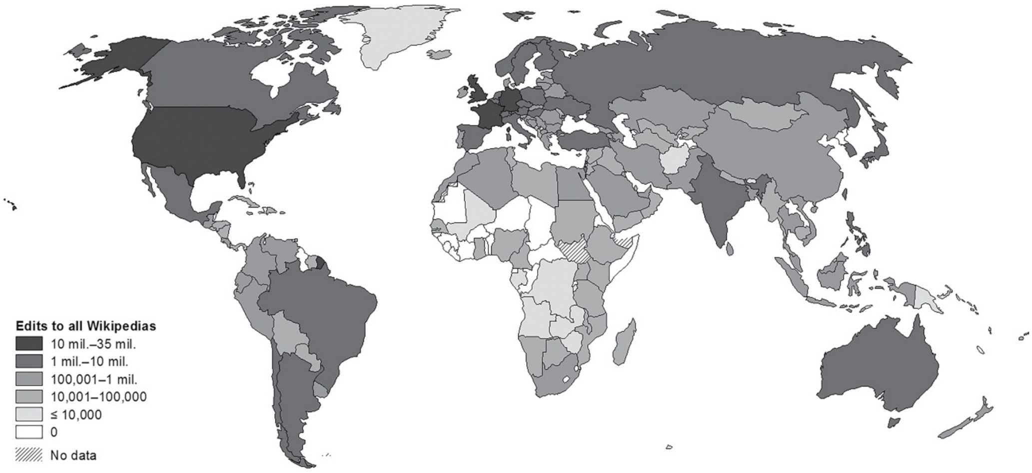


Figure 1: Distribution of edits to all language editions of Wikipedia (2013) [6]

in the stock of knowledge in the respective economies". In other words, availability of knowledge in each economy can impact how much people engage with Wikipedia. This connects back to Graham et al.'s [6] research, where they established how broadband internet connection is a central predictor of uneven geographic participation. Converging evidence from both studies highlights that high-income countries generally have more Wikipedia clicks and contributions. However, they also reveal outliers, where usage patterns diverge significantly despite large internet user bases.

2.3 Bias in Knowledge Production and Representation

Building on previous findings, recent studies extend the understanding of inequality in Wikipedia. In [18], Mittermeier et al. point out that language serves as the best proxy for understanding geographic patterns of Wikipedia use, for location information isn't a part of the metadata provided by the platform's API. In fact, language is a more reliable surrogate for region-specific languages, for example those spoken in northern or eastern Europe. Furthermore, languages reflect Wikipedia's geographical bias. Although the majority of Wikipedia languages are European, the diversity of spoken languages is highest in Asia and Africa [18] [19]. The only exception, again, is China, where access to Wikipedia and other internet platforms has been restricted.

Beytía [7] argues how the geographical bias in multilingual Wikipedia shouldn't only consider the number of articles tied to specific locations, but also their internal positioning. Moreover, the same content may not be available or equally detailed across different language editions of Wikipedia. As an example, a historical site in one country might have a detailed article in the local language, but a very brief entry in another one. The internal positioning also includes the centrality in the network of references, which refers to how often articles are linked or referenced by other articles. Beytía [7] studies this idea through empirical research that systematically analyses the geographic concentration in Wikipedia's biographical coverage of globally recognised individuals—those with biographies available in over 25 languages.

As a result, he reveals that the distribution of BCI (Biographical Centrality Index) is mainly concentrated in the Global North, specifically in the United States of America, the United Kingdom, Italy, France and Germany. As a matter of fact, the concentration of the biographical coverage is so pronounced that these countries account for 50.6% of all biographies available in over 25 languages. This concentration increases to 62.1% when factoring in the articles' positioning [7]. In addition, the results revealed that a similar pattern emerges when comparing the number of biographies of individuals born in different countries, a Gini coefficient of 0.79 and a Palma ratio of 41 are observed. This indicates that the top 10% of countries with the highest biographical coverage have 41 times the number of biographies compared to the bottom 40%. The Gini coefficient and Palma ratio increase

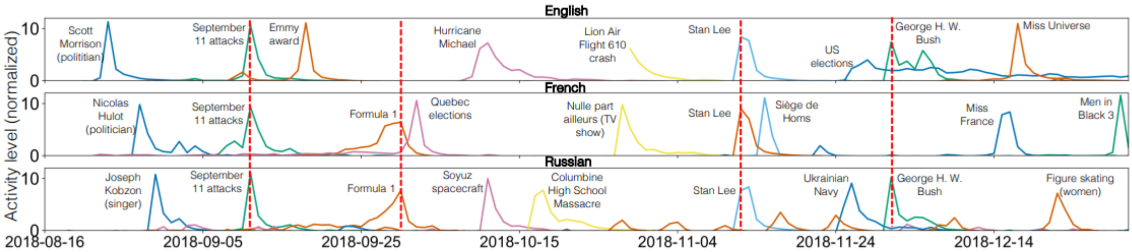


Figure 2: Trends over time and across different language editions are represented by colored curves, each linked to a specific trend with a brief description [20]

even further when taking into account the positioning of articles, .84 and 207 respectively.

Voldymir Miz, et al. [20] focus on the English, French and Russian Wikipedia editions to identify and compare the most popular topics among the readers over the last 4 months of 2018. The authors first identify the trends by extracting a sub-network of popular Wikipedia articles. They then analyse summaries of the articles in the sub-network to extract and prioritise keywords based on their significance. Lastly, they assign high-level topics to the identified trends using the extracted keywords. The results differ based on the trends and events throughout the 4 months. There are events that appear in all languages, and events that appear only in a few of them. For instance, the 09/11 commemoration appears in all of the editions, while Formula 1 appears in only 2 of the 3 languages, French and Russian (Figure 2). If an event is local or related to a specific culture of the country, for instance, the death of a popular Russian singer, it is likely to not grab the attention of other Wikipedia regions.

According to the authors, the reason why some topics are different across all languages is linked with media coverage, geographic proximity and cultural differences. It is confirmed that media inflates people's interest in the trends. On average, approximately 25% of readers are driven by media coverage [20] [11] [10]; for English and Russian versions this can rise up to 30%. Geographic proximity plays a big role as well, particularly in the case of natural disasters. Traumatic events usually spread across all languages, but a hurricane in North Carolina would not spark the same interest it does in French and Russian versions that it would in the English edition.

3 Project Specification

3.1 Motivation:

Wikipedia is an ever-growing platform, which consists of vast knowledge in many different languages of the world. The accessibility of it makes it possible for people to gather new knowledge and self-develop academically. Unfortunately, high-income countries are overrepresented in the knowledge production on Wikipedia, while low-income countries barely contribute to the platform and have their own voice. My main motivation of doing this project stems from the need to address inequities in the global knowledge ecosystem. I want to investigate the disparities in contributor demographics and content coverage to unveil deeper issues of representation and access.

3.2 Aims and Objectives:

- Analyse patterns of contributions to Wikipedia by focusing on geographic, cultural, and socioeconomic disparities.
- Investigate the consumption of Wikipedia knowledge by examining page view data across all available languages and regions.
- Identify biases in content representation and analyse them.

3.3 Scope:

This project will focus on the following areas:

- Contributor patterns: Who edits Wikipedia? Regional and demographic biases.
- Content representation: Geographic distribution and coverage bias in the Wikipedia articles.
- Consumption behaviour: Analysis of user page views across different languages, topics and regions.
- Knowledge gaps: Identification of areas with low content engagement and activity.

3.4 Methodology:

3.4.1 Datasets and preprocessing

SQL will be used for large-scale data extraction and management. I will first extract the data using Wikimedia APIs (MediaWiki API, Meta-Wiki API, Wikimedia REST API, and so on) and dumps, and then I will process the user edits and page views data by focusing on user activities such as timestamps, article edits, and page views broken down by region and time. Missing or null values will be removed to handle inconsistencies. Additionally, the data might have irrelevant fields like bot edits which will need to be removed as well. Other data processing techniques (data reduction, enrichment, validation, etc.) will be applied wherever necessary. Python will be the main programming language for preprocessing. Possible libraries include Requests, Pandas, NumPy, Matplotlib, Scikit-learn, TensorFlow.

For external data on socioeconomic factors I will go through UN datasets and extract the data in a similar fashion by aligning metrics like GDP, internet access, and literacy rates with Wikipedia's data to standardise formats and remove any anomalies. Once both of the datasets are processed and managed appropriately, I will integrate them using shared keys such as country or region.

3.4.2 Data mining, network and spatial analysis

Data mining and network analysis will be done by analysing the relationships between editors, articles and topics using network science. Moreover, I will visualise the content gaps and contributor networks with the help of either Gephi or NetworkX. Geographic biases and the visualisation of global content distributions will be mapped with GIS tools.

3.4.3 Machine learning

The identification of underrepresented regions will be accomplished using predictive models. I will use activity and preferences for clustering all editors.

3.5 Deliverables

3.5.1 Data processing and analysis outputs

- Cleaned datasets: Preprocessed and structured data from Wikimedia sources and UN datasets.
- Data insights reports: Statistical summaries and visualisations of patterns in Wikipedia editor and reader trends.

3.5.2 Geographic and socioeconomic analysis

- Bias maps: Geographic visualisations showing disparities in Wikipedia content contributions and consumption.
- Correlations: Analytical reports which highlight the relationships between socioeconomic factors and Wikipedia activity.

3.5.3 Machine learning models

- Behaviour predictions: Trained models that predict user editing or viewing behaviours based on available data.

3.5.4 Interactive dashboards

- Dynamic tools which will display live or preprocessed visualisations of edit, page view, and content trends.

3.5.5 Final report

- Documentation of findings, methodologies, and all outcomes of the project.

4 Evaluation Criteria

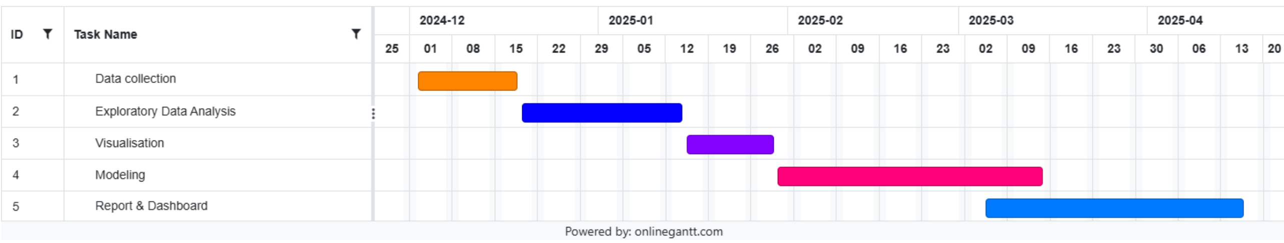
The proposed project should be evaluated based on the following evaluation criterias.

Evaluation Criteria	Description
Data quality	The data presented must be complete, accurate, and consistent. The integration of external data, such as UN dataset and Wikimedia data, must be smooth and not have any delays in collection or formatting issues.
Analysis and insight	The identified trends should be relevant and preserve clarity. Quality of correlations between socioeconomic factors and Wikipedia activity should be considered as well.
Model performance	The model's performance should be accurate and reliable. These should be measured using metrics like accuracy, precision, recall, etc.
Visualisation and communication	The maps, charts, and dashboards should preserve clarity and should be accessible. The final report must convey results effectively.
Ethical considerations	All the data must adhere to privacy and ethical standards of handling user-related data.

Table 1: Evaluation Criteria and Description

5 Project Timeline

This timeline provides a simple overview of relevant dates and key milestones in my project. Some of the milestone dates could change a little, but this shouldn't affect the overall big picture of the timeline.



6 Conclusions

To conclude this paper, we first discussed the relevant literature, divided into three parts: Overview of readers and editors in Wikipedia, Geographic distributions of Wikipedia readers and editors, and Bias in knowledge production and representation. In the first part we reviewed the main motivators of readers and editors to understand the reason why they contribute and consume Wikipedia articles. The biggest motivator for both was intrinsic motivation; however, we also revealed that intrinsic motivation does not have any effect on the knowledge-sharing behaviour of the editors since they enjoy the process of editing more, and thus are more motivated by internal self-concept motivation.

After analysing geographic distributions of Wikipedia readers and editors, we came to a conclusion that Wikipedia edits are primarily concentrated in economically developed countries, such as the United States of America, Germany, and Japan. Although it seemed like country's wealth would be the determining factor for edit distributions, it was actually Broadband Internet Access. Countries with either low or high BCIs contributed to Wikipedia more than expected. However, there were outlier countries with restricted access to the internet like China and Iran.

Furthermore, language was highlighted as a proxy for geographic patterns. Most Wikipedia languages are European; however, Asia and Africa are considered to be more linguistically diverse. Additionally, the concentration of biographical coverage is highest in the Global North (e.g., USA, UK, Germany). Overall, we found out that the popularity of topics is mainly influenced by media coverage, geographic proximity, and cultural differences.

In the last section, project specification, we covered the motivation for this project, as well as aims and objectives, scope, methodology, deliverables, timeline, and evaluation criteria. Future work could explore the Wikipedia inequality further by focusing more on qualitative data by conducting interviews or surveys with contributors to understand them, and assess the participation bias from a sociological viewpoint.

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